

DRISHYAM

SIH26189 Change Roadmap

File-level change plan to convert the existing DRISHYAM platform
into an AI-Powered Criminal Network Analysis System

Problem Statement: SIH26189 — AI-Powered Criminal Network Analysis System

Organisation: Ministry of Home Affairs · NCRB, Women Safety Division

Repository audited: `Drishyam_Latest_Integrated` (backend + frontend)

Basis: Direct file-level source audit, cross-checked against three advisory reports

Prepared for: the DRISHYAM SIH Project Team

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1. Executive Summary — what actually needs to change

Core finding: DRISHYAM already contains roughly 60% of what SIH26189 requires — but it is dressed for the wrong problem. The evidence-provenance engine is strong and should not be rebuilt. What must change is the **entity vocabulary**, the **shape of the graph**, and the **presentation layer**.

The most important architectural principle recommended by the advisory reports — Evidence → Observation → Claim → Entity/Event/Relationship → Candidate or Verified Graph — **is already implemented** in this repository. It exists as the chain RawExtractionArtifact → ModelInferenceRun → NormalizedRecord → RecordRelation → RecordReview. The advisory reports could not confirm this because they were written without access to the source repository. This audit confirms it from the code.

The consequence for planning is significant:

- **No rewrite is required.** The provenance, review, audit, storage, worker and model-gateway layers are production-shaped and should be retained as-is.
- **The real work is additive.** Three additions unlock everything else: a criminal-network entity vocabulary, an entity-to-entity relationship table, and network analytics.
- **The current system is tuned for cyber-fraud, not criminal networks.** Its extractors, alert rules and graph shape all assume UPI/phishing investigation. SIH26189 asks for people, vehicles, locations, organisations and their relationship maps.

THE THREE CHANGES THAT MATTER MOST

#	CHANGE	WHY IT IS DECISIVE	EFFORT
1	Add Person / Vehicle / Location / Organisation extraction	The problem statement names these five entity classes explicitly. Vehicles, locations and organisations are currently not extracted at all . This is the first thing an evaluator will look for.	2–3 days

2	Add an <code>entity_relations</code> table with typed, provenance-bearing edges	The current graph connects <i>evidence files</i> to <i>shared identifiers</i> . A criminal network graph connects <i>entities</i> to <i>entities</i> . Entity-to-entity edges do not exist today.	3–4 days
3	Add NetworkX centrality, bridge and community analytics	"Identify influential individuals" is an explicit requirement with zero current implementation. NetworkX is already an installed dependency, so this is the cheapest large win.	2 days

2. How this document was produced

This roadmap is based on a direct file-level audit of the repository, not on the advisory PDFs alone. The following were read in full or in relevant part:

AREA	FILES EXAMINED
Application entry & routing	<code>app/main.py</code> , <code>app/api/router.py</code> , all 12 route modules (~68 endpoints)
Data model	<code>app/models/entities.py</code> (804 lines, 33 tables)
Configuration & security posture	<code>app/core/config.py</code> , <code>app/core/security.py</code>
Processing pipelines	<code>app/services/pipeline.py</code> , <code>app/services/grounded_pipeline.py</code> , <code>app/workers/tasks.py</code>
Evidence intelligence	<code>app/evidence_intelligence/</code> — schema, extraction, ocr, providers/router, correlation, confidence
Graph & resolution	<code>app/graph/connections.py</code> , <code>app/graph/projection.py</code> , <code>app/services/entity_resolution.py</code> , <code>app/services/cross_case.py</code>
Extraction & rules	<code>app/extraction/normalizer.py</code> , <code>app/alerts/rules.py</code> , <code>app/parsers/registry.py</code>
Frontend	<code>client/src/App.tsx</code> , <code>pages/Workspace.tsx</code> , <code>pages/workspace/*</code> , <code>api/*</code>
Deployment	<code>docker-compose.yml</code> , <code>requirements.txt</code> , <code>package.json</code> , <code>vite.config.ts</code>

Where this document disagrees with the advisory reports, the disagreement is stated explicitly in Section 5 with the reason.

3. Current state audit — what already exists and must be reused

The following capabilities are **already implemented and verified in code**. They must be retained. Rebuilding any of them would consume the schedule without improving the evaluation outcome.

CAPABILITY	STATUS	WHERE IT LIVES
Original evidence preserved; SHA-256 integrity receipt	Complete	<code>services/storage.py</code> , <code>services/trustify.py</code> , <code>TrustifyReceipt</code>
Asynchronous processing, five dedicated Celery queues	Complete	<code>workers/celery_app.py</code> , <code>docker-compose.yml</code>
Observation → Claim → Review lifecycle	Complete	<code>models/entities.py</code> — <code>RawExtractionArtifact</code> , <code>NormalizedRecord</code> , <code>RecordRelation</code> , <code>RecordReview</code>

Field-level provenance: value + basis + literal quote + source reference + reason	Complete	evidence_intelligence/schema.py — FieldProvenance
"Never guess" contract — null is a first-class value carrying a reason	Complete	FieldProvenance.unestablished()
Model may never assert guilt — enforced at validator level	Complete	schema.py — FORBIDDEN_EVENT_TYPES
Provider-neutral model gateway (Ollama / Groq / mock)	Complete	evidence_intelligence/providers/router.py , providers/base.py
External evidence transmission is double-gated and off by default	Complete	core/config.py — groq_transmission_allowed
Assistant is architecturally isolated from case data	Complete	Separate credential block in config.py
OCR producing bounding boxes and quality flags	Complete	evidence_intelligence/ocr.py — pytesseract.image_to_data
Human review queue with append-only correction history	Complete	RecordReview , api/grounded.py
Entity resolution — one identifier, one node, many sightings	Complete	services/entity_resolution.py , EntityOccurrence
Cross-case identifier matching, authorisation-scoped	Complete	services/cross_case.py
Explainable, idempotent alert rules	Complete but wrong domain	alerts/rules.py — 5 rules
Audit log, RBAC, case-level authorisation	Complete	AuditLog , CaseMembership , api/deps.py
Idempotent reprocessing via processing-run IDs	Complete	ProcessingRun , unique idempotency_key throughout
Server-side report generation	Complete	services/reporting.py (1173 lines)

Planning consequence. The "audit and stabilise" and "common data model" phases recommended by the advisory reports are already satisfied. The team can start at what those reports call Phase 3, which saves roughly two weeks of the proposed schedule.

4. Gap analysis against SIH26189

SIH26189 requires a system that processes multiple crime and intelligence sources; extracts **people, locations, vehicles, phone numbers and organisations**; builds relationship maps; identifies influential individuals; detects suspicious patterns; and provides visual and analytical insight for investigators. Each obligation is assessed below.

PS OBLIGATION	STATUS	FINDING
Process multiple sources	Partial	Generic parsers exist for TXT, EML, CSV, XLSX, PDF and images. There is no CDR adapter and no FIR/police-report adapter — the two sources the problem statement names first. <code>SourceType</code> has no <code>FIR</code> , <code>CDR</code> , <code>SURVEILLANCE</code> or <code>SOCIAL_MEDIA</code> member.
Extract entities	Major gap	Current vocabulary (<code>extraction/normalizer.py</code>) is: <code>phone</code> , <code>upi_id</code> , <code>email</code> , <code>url</code> , <code>ifsc</code> , <code>account_number</code> , <code>utr</code> , <code>ip_address</code> , <code>person</code> . Vehicle: absent. Location: not an entity (only a plain string field on the record). Organisation: absent. Person extraction requires a literal <code>Name: / Beneficiary:</code> prefix, so it will not fire on free-text FIR narrative.
Build relationship maps	Major gap	<code>graph/connections.py</code> builds an evidence ↔ identifier bipartite graph — it answers "what do these two files have in common". SIH26189 needs entity → entity edges. <code>RecordRelation</code> only carries <code>corroboration / contradiction</code> between <code>records</code> . There is no <code>CALLED</code> , <code>TRANSFERRED_TO</code> , <code>LOCATED_AT</code> or <code>USED_VEHICLE</code> anywhere in the codebase.
Identify influential individuals	Absent	A repository-wide search for <code>centrality</code> , <code>betweenness</code> , <code>pagerank</code> , <code>community</code> and <code>louvain</code> returns zero results . <code>NetworkX</code> is already installed and used only for a provenance projection.
Detect suspicious patterns	Wrong domain	The five existing rules are <code>HIGH_VALUE_TRANSFER</code> , <code>REPEATED_RECIPIENT</code> , <code>RECURRING_IDENTIFIER</code> , <code>PHISHING_LINK</code> , <code>RAPID_TRANSACTION_SEQUENCE</code> . These are payment-fraud rules. They are well built — explainable, idempotent, evidence-linked — but they do not address criminal-network patterns.
Temporal analysis	Absent	No incident window on the <code>Case</code> model, no before/after analysis, no communication-burst detection. The timeline is a chronologically sorted list only.
Visual / analytical insight	Partial	Sixteen workspace views exist, but there is no Entity Profile screen , no Relationship Detail drill-down , and no graph filtering, hop-expansion or path highlighting. The differentiating chain <code>graph → relationship → claim → evidence → original source</code> cannot currently be clicked through in the UI.
Actionable intelligence	Partial	The Trace Orb assistant is deliberately and correctly isolated from case data. A separate case-grounded, permission-aware, source-citing assistant does not exist.

Non-functional gaps

ITEM	SEVERITY	FINDING AND REQUIRED ACTION
Problem statement number	Medium	<code>Readme.md</code> states " inspired by SIH25198 ". The team is submitting to SIH26189 . Product framing, capability language and the MHA / NCRB Women Safety Division context must be corrected throughout.

Version control state	High	The repository has a single commit . The entire <code>app/evidence_intelligence/ subsystem</code> , <code>api/grounded.py</code> , <code>graph/connections.py</code> and <code>schemas/grounded.py</code> are untracked , alongside modifications to fourteen tracked files. A single mistaken checkout would destroy the strongest part of the project. There is also no <code>develop</code> branch and no feature branches, which contradicts the contribution rules the project's own README defines.
Secret handling	High	<code>backend/.env</code> holds live provider credentials on disk inside the repository folder. It is correctly git-ignored and untracked, but the README itself instructs storing it outside the repository at <code>~/.drishyam/</code> . That instruction is not being followed.
Residual case data	Medium	<code>backend/data/private_storage/</code> and <code>backend/backups/*.sql</code> contain real case artefacts and a database dump. These must be cleared before any demonstration or repository publication.
Deployment expectation	Medium	<code>docker-compose.yml</code> defines only <code>api</code> and <code>worker</code> — no PostgreSQL, no Redis. Both are external services, so the stack cannot be brought up by a judge or a new team member without the private environment file. A self-contained demo profile is required.

5. Explicit non-goals — what we are deliberately NOT building

The advisory reports recommend several additions that this audit rejects on the basis of what the repository actually contains. Each rejection is stated with its reason so the decision can be defended if questioned.

We will not add Neo4j.

Two of the advisory reports suggest Neo4j as a graph projection. The expected case graph is a few thousand nodes. NetworkX is already an installed dependency and is sufficient at that scale. Adding Neo4j means another container, another migration path, another failure mode, and no visible benefit in a demonstration. One of the reports itself concedes "NetworkX for MVP". We follow that. PostgreSQL remains the authoritative system of record either way.

We will not switch OCR engines to PaddleOCR.

The stated reason for PaddleOCR is bounding boxes and layout coordinates. The existing `pytesseract.image_to_data` implementation in `evidence_intelligence/ocr.py` *already produces bounding boxes*, block-level confidence and quality flags. PaddlePaddle wheels are large and version-sensitive and would put the Docker build at risk close to the event. The real accuracy improvement comes from **OpenCV preprocessing** applied before OCR — that is what we will add instead.

We will not rewrite the model gateway for OpenAI.

One report specifies building a `VisionExtractionService` interface. That interface already exists as `ModelRouter` with a `providers/base.py` protocol and three implementations. The correct change is to add one file, `providers/openai.py`, conforming to the existing protocol — approximately 120 lines. Nothing above the provider layer changes.

We will not hard-code any model identifier.

Model names must continue to arrive from environment variables, exactly as `OLLAMA_VISION_MODEL` and `GROQ_MODEL` do today. Model identifiers quoted inside advisory documents must be verified against the live provider catalogue before use and must never be embedded in source.

We will not connect the Trace Orb assistant to case data.

Trace Orb is intentionally isolated: it uses a separate credential block so that a website help assistant can never reach evidence. This is a correct safety boundary and a defensible design point in evaluation. The case-grounded assistant will be a **separate** service reading only database rows.

We will not add blockchain, Kubernetes, Kafka, a service mesh or a vector database.

All three advisory reports agree, and this audit agrees. Hash chains, lineage records, audit logs and access control already provide the integrity story. Infrastructure added for appearance costs schedule and invites questions the team cannot answer with measurements.

6. Data model changes

The schema changes required are small, because the provenance model is already correct. Three of the four changes need no migration at all.

CHANGE	MIGRATION?	DETAIL
New entity types (vehicle, location, organisation)	Not required	Entity.entity_type is String(64) , free-form. New types need extraction code only. Weights are added to IDENTIFIER_WEIGHT in graph/connections.py and to IDENTITY_FIELDS in services/entity_resolution.py .
New source types	Enum values only	SourceType in schema.py gains FIR , POLICE_REPORT , CDR , SURVEILLANCE , SOCIAL_MEDIA . It is a Python str enum persisted as text, so no type migration is needed.
New record fields	Small	normalized_records gains vehicle_identifiers (JSON), organisations (JSON), and a structured location_detail (JSON). One additive Alembic revision.
New table: entity_relations	Required	The single structural addition. Defined below.
Incident window on Case	Small	cases gains incident_window_start and incident_window_end (nullable timestamps). Anchors all temporal rules.

The `entity_relations` table

```
class EntityRelation(Base):
    """A typed, provenance-bearing edge between two resolved entities.

    Invariant: an edge is never inserted without source_evidence_id and
    source_reference. A relationship without provenance is not investigative
    intelligence and must not reach the graph.
    """
    __tablename__ = "entity_relations"

    id : str # uuid
    case_id : FK cases.id (indexed, CASCADE)

    subject_entity_id : FK entities.id (indexed)
    object_entity_id : FK entities.id (indexed)
    relation_type : String(32) # CALLED | MESSAGED | TRANSFERRED_TO
    # LOCATED_AT | USED_VEHICLE
    # MENTIONED_WITH | ASSOCIATED_WITH_CASE

    source_evidence_id : FK evidence_files.id (NOT NULL)
    source_record_id : FK normalized_records.id (nullable)
    source_reference : JSON # bbox, page, or CSV row (NOT NULL)

    observed_at : DateTime | None
    time_precision : String(16) # reuse TimePrecision vocabulary
    support_count : Integer # independent records supporting this edge
    supporting_record_ids : JSON

    extraction_method : String(64)
    confidence : Numeric(5,4)
    verification_status : String(32) # machine_extracted | needs_review
    # human_verified | rejected | superseded

    reviewed_by_id : FK users.id | None
    reviewed_at : DateTime | None
    idempotency_key : String(255) UNIQUE
    created_at : DateTime

    __table_args__ = (
        Index("ix_entity_relations_case_type", "case_id", "relation_type"),
        Index("ix_entity_relations_subject", "case_id", "subject_entity_id"),
        Index("ix_entity_relations_object", "case_id", "object_entity_id"),
        Index("ix_entity_relations_time", "case_id", "observed_at"),
    )
```

Why the unique `idempotency_key` matters. Reprocessing the same evidence must not duplicate edges. The key is derived from `case_id + relation_type + subject + object + source_record_id`. This mirrors the pattern already used in `alerts/rules.py` and `RecordRelation`, so it is consistent with the existing codebase.

7. The roadmap — Phase 0 to Phase 9

Phases are ordered by **demonstration visibility**, not by architectural tidiness. The system remains demonstrable after every phase. Each phase states exactly what changes and what the change produces.

Phase 0 — Repository hygiene and correct positioning

Effort: 2–4 hours · Priority: do this before writing any code

WHAT CHANGES

1. Commit all untracked work. Create `develop`, then four feature branches as the project README already prescribes.
2. Move `backend/.env` to `~/.drishyam/drishyam-staging.env` and set `DRISHYAM_RUNTIME_ENV_FILE`. Rotate any credential that has been shared outside the team.
3. Clear `backend/data/private_storage/`, `backend/backups/*.sql` and `generated_reports/`.
4. Rewrite `Readme.md`: `SIH25198` → `SIH26189`; add MHA / NCRB Women Safety Division framing; restate the product as criminal-network intelligence rather than cybercrime evidence.
5. Add a `docker-compose.demo.yml` that includes PostgreSQL and Redis so the stack can be started by anyone without the private environment file.

Result: the strongest part of the codebase stops being one command away from deletion; the repository reads as a SIH26189 submission; and any evaluator or new team member can start the stack from a clean clone.

Phase 1 — Criminal-network entity vocabulary

Effort: 2–3 days · Highest return on effort · No structural migration

WHAT CHANGES

FILE	CHANGE
<code>app/extraction/entities_in.py</code> (new)	Vehicle — Indian registration plates: <code>[A-Z]{2}\s?\d{1,2}\s?[A-Z]{1,3}\s?\d{4}</code> plus BH-series. Deterministic and high-confidence. Location — gazetteer-driven: a curated list of districts and police-station names combined with context words (<code>at</code>, <code>near</code>, <code>PS</code>, <code>Thana</code>). Open-domain NER is deliberately avoided: it is unreliable on Indian place names and would flood the review queue. Organisation — suffix-driven: <code>Pvt Ltd</code>, <code>Bank</code>, <code>Enterprises</code>, <code>Traders</code>, & <code>Sons</code>. Person — two channels: the existing labelled regex, plus vision-model extraction recorded with <code>basis=DIRECT_VISUAL</code>. Both set <code>requires_human_review=True</code>.
<code>evidence_intelligence/schema.py</code>	Add <code>vehicle_identifiers</code> , <code>organisations</code> , <code>location_detail</code> to <code>NormalizedRecordDraft</code> . Extend <code>SourceType</code> with <code>FIR</code> , <code>POLICE_REPORT</code> , <code>CDR</code> , <code>SURVEILLANCE</code> , <code>SOCIAL_MEDIA</code> .

<code>services/entity_resolution.py</code>	Add the three new fields to <code>IDENTITY_FIELDS</code> . Extend <code>canonicalize()</code> : plate normalisation strips spaces and hyphens and upper-cases; organisation normalisation strips legal suffixes before comparison.
<code>graph/connections.py</code>	Add weights to <code>IDENTIFIER_WEIGHT</code> : <code>vehicle: 0.90</code> (a plate is near-unique), <code>organisation: 0.50</code> , <code>location: 0.35</code> (many people share a location; it is a weak link).
<code>evidence_intelligence/providers/prompts.py</code>	Extend the extraction prompt to request vehicles, locations and organisations, retaining the existing literal-transcription and null-when-unreadable contract.

Result: uploading a single FIR PDF now yields Person, Vehicle, Location, Phone and Organisation entities, each carrying a source reference and a confidence band. The system visibly answers the entity-extraction requirement of the problem statement, which it currently cannot.

Phase 2 — Typed entity-to-entity relationships

Effort: 3–4 days · The structural centrepiece of this roadmap

WHAT CHANGES

1. Add the `EntityRelation` model (Section 6) and one Alembic revision.
2. New service `app/services/relationship_builder.py` deriving edges from `NormalizedRecord` rows:

SOURCE RECORD	EDGE PRODUCED
CDR row	Phone —CALLED→ Phone with <code>observed_at</code> and duration
Transaction row	Account —TRANSFERRED_TO→ Account
Chat record	Participant —MESSAGED→ Participant, only when the source establishes the role — the existing <code>chat_participant_identifier</code> discipline is preserved
FIR / surveillance record	Person —MENTIONED_WITH→ Vehicle, Person —LOCATED_AT→ Location, Person —USED_VEHICLE→ Vehicle

3. Call the builder from `services/grounded_pipeline.py` after entity resolution, inside the existing idempotent processing-run boundary.
4. New endpoints in `api/grounded.py`: `GET /cases/{id}/entity-relations` (filterable by type, date, verification status) and `POST /entity-relations/{id}/review` reusing the existing review flow.

Non-negotiable invariant. Every edge must carry `source_evidence_id` and `source_reference`. An edge without provenance is never inserted. Enforcing this at insertion time is what makes the project's headline metric — Traceability Coverage — provably 100%, rather than an estimate.

Result: DRISHYAM produces an actual criminal-network graph rather than a file-similarity graph. Every edge can be traced to the file, page, row or pixel region it came from. This single change converts the product from an evidence organiser into a network-analysis system.

Phase 3 — Network analytics: influential individuals

WHAT CHANGES

- New module `app/graph/analytics.py` building a `networkx.Graph` from `entity_relations` and computing: degree centrality, betweenness centrality, PageRank, `nx.bridges`, `louvain_communities`, shortest path between two entities, and `k`-hop subgraph extraction for progressive UI loading.
- New endpoints: `GET /cases/{id}/network/important`, `GET /cases/{id}/network/subgraph`, `GET /cases/{id}/network/path`.
- Each ranked entity is returned with a generated **explanation string**, never a bare score.

```
{
  "entity": "98XXXXXX21",
  "entity_type": "phone",
  "metric": "betweenness_centrality",
  "rank": 1,
  "why": "This phone connects 3 otherwise separate groups in this case and
        appears in 7 records across 4 evidence files.",
  "supporting_evidence_ids": ["ev-1", "ev-4", "ev-7", "ev-9"],
  "caveat": "Network importance indicates review priority.
            It is not an indication of guilt."
}
```

Result: the "identify influential individuals" requirement is satisfied with an explanation an investigator can act on and a judge can interrogate — not an unexplained score, which the advisory reports and this audit both identify as the primary interpretive risk.

Phase 4 — Source adapters for the problem statement's named sources

Effort: 3 days

WHAT CHANGES

Adapters are added to `app/parsers/` in priority order. All emit the existing `NormalizedRecord` contract. **No adapter is permitted to touch graph logic** — this is what prevents CSV, OCR and vision parsers from producing incompatible representations of the same fact.

#	ADAPTER	EXTRACTS
1	CDR (CSV/JSON)	Header auto-detection for A-party, B-party, date, time, duration, cell ID, IMEI/IMSI. Phone normalisation reuses the existing <code>normalize()</code> . Emits communication events and <code>CALLED</code> edges.
2	FIR / police report (PDF, scan)	FIR number, sections invoked, date, police station, complainant, accused, vehicles, locations, narrative events. PDF text layer first; OCR fallback already exists.
3	Financial transaction (CSV/JSON)	Largely present. Extended to emit <code>TRANSFERRED_TO</code> edges with participants, amount, reference and timestamp.
4	Surveillance note (PDF/text/image)	Persons, vehicles, locations, observation times from free-text narrative.
5	Social-media export (JSON/CSV/screenshot)	Handles, posts, mentions, URLs, timestamps. Lowest priority; include only if schedule allows.

Result: a single case can be built from four or more heterogeneous source types — the explicit demonstration requirement of the problem statement — with the same fictional entities appearing across several of them, which is what makes cross-source resolution visible.

Phase 5 — Temporal intelligence and criminal-network alert rules

Effort: 2–3 days

WHAT CHANGES

- Add `incident_window_start` / `incident_window_end` to `Case`; expose them in case creation and the workspace UI.
- New module `app/services/temporal.py`: event sequencing, before/after analysis relative to the incident window, communication-burst detection, and cross-source event correlation.
- Extend `alerts/rules.py`. Existing fraud rules are retained but no longer dominant. Add a `rule_version` field to every alert.

NEW RULE CODE	DETECTS
<code>PRE_INCIDENT_COMMUNICATION</code>	Repeated contact between two entities in the N hours preceding the incident window
<code>SHARED_IDENTIFIER_ACROSS_CASES</code>	One phone, account or vehicle appearing in multiple authorised cases — reuses <code>services/cross_case.py</code>
<code>RECURRING_VEHICLE_AT_LOCATION</code>	The same registration plate observed at the same location across multiple sightings
<code>COMMUNICATION_BURST</code>	A statistically unusual spike in call or message frequency within a short window
<code>BRIDGE_ENTITY</code>	High betweenness centrality — fed directly from Phase 3

Result: alerts speak the language of criminal-network investigation rather than payment fraud. Each remains explainable, versioned, idempotent and linked to the evidence that produced it — an investigation lead, never a legal conclusion.

Phase 6 — Vision upgrade: preprocessing, quality routing, OpenAI provider

Effort: 3 days · Incremental addition, not a replacement

WHAT CHANGES

1. **OpenCV preprocessing** — new `evidence_intelligence/preprocessing.py`: deskew, denoise, upscale, adaptive and Otsu thresholding, contrast adjustment, region cropping. Every variant is stored as a **derived artefact linked to the parent evidence ID**. The original is never overwritten — a discipline the codebase already observes.
2. **Quality report and routing** — resolution, Laplacian-variance blur estimate, contrast, OCR coverage, producing a recommendation of `ocr_only`, `vision_review` or `human_review_required`. This is what prevents an expensive vision call on every clear printed page.
3. **OpenAI provider** — new `providers/openai.py` implementing the existing `providers/base.py` protocol, using Structured Outputs with a strict JSON schema. Add `openai_first` to `LLM_ROUTING_MODE`. The model identifier arrives from `OPENAI_VISION_MODEL` and is verified against the live catalogue before use.

4. **Handwriting states** — add a `readability` field with values `readable`, `partially_readable`, `unreadable`, alongside a literal transcription. A partially readable value is **never normalised into a complete one**: a fragment such as `98?7?2` must never become a phone number or a graph node.
5. **Response cache** keyed on `sha256 + preprocessing_version + model_id + prompt_version`, so an unchanged image is never sent twice.

Result: handwritten and low-quality evidence is handled honestly. A blurry FIR page produces "unclear — review required" with the visible fragment preserved, rather than an invented value. In evaluation this is the strongest safety demonstration available, and it costs nothing to show.

Phase 7 — Investigator workspace

Effort: 4–5 days · Frontend · Can run in parallel with Phases 2–5

WHAT CHANGES

SCREEN	REQUIREMENT
Network Intelligence upgrade of Entities & Graph	Filters by entity type, relation type, date range, source and confidence; one- and two-hop expansion; shortest-path highlighting; a legend; and visually distinct candidate versus verified edges . The dashed/solid convention already exists in <code>graph/connections.py</code> and should be carried through.
Entity Profile (new)	Aliases, normalised identifiers, connections, events, cases the entity appears in, source list, resolution status and verification status.
Relationship Detail (new)	Relation type, support count, confidence broken down by dimension, source excerpt, source file, and a bounding-box overlay on the original image .
Timeline upgrade	Incident-window filter, before/after view, entity and source filters.

TWO RULES THAT GOVERN THE WHOLE UI

- **An unreviewed machine-extracted edge must never look identical to a human-verified edge.** This is the single strongest shared recommendation across all three advisory reports and it is a product-safety requirement, not a styling preference.
- **The browser must never load the whole graph.** The frontend requests an authorised, filtered subgraph and expands progressively.

Result: the differentiating chain becomes clickable end to end — `graph` → `relationship` → `claim` → `evidence reference` → `original source`. Most competing entries will show a graph; very few can click from a graph edge to the original pixel region it came from. The provenance data required for this already exists in `FieldProvenance.source_reference` and the OCR bounding boxes; only the interface is missing.

Phase 8 — Case-grounded assistant

Effort: 2 days · Trace Orb remains untouched

WHAT CHANGES

- New service `app/services/case_assistant.py`, entirely separate from the Trace Orb help assistant.

- It reads **only database rows** — entities, relations, records, alerts, audit entries. It must never use general model knowledge to fill a missing case fact.
- It reuses the existing case-authorisation dependency in `api/deps.py` ; permissions are enforced before any data is retrieved, not after.
- Every factual answer returns source references, verification status and an explicit uncertainty statement.
- All uploaded document text is treated as **untrusted data**. Prompt injection inside an uploaded file must not be able to change extraction policy, call tools, alter permissions or instruct the model to ignore uncertainty. This is tested adversarially, not assumed.

Result: a question such as *"What evidence connects Person A with the vehicle in this case?"* is answered with citations, verification states and, where appropriate, an explicit "not established by the evidence" — while the website help assistant remains provably unable to reach case data.

Phase 9 — Benchmark dataset and measurement

Effort: 3 days

WHAT CHANGES

Extend the existing `scripts/generate_synthetic_evidence.py` to produce one realistic synthetic case:

- 2 FIR / police-report PDFs · 3 WhatsApp screenshots (one dark mode, one deliberately blurry) · 1 CDR CSV · 1 transaction CSV · 1 surveillance note
- **The same fictional entities must appear across several sources** — without this, cross-source resolution cannot be demonstrated at all
- Deliberate hard cases: one unreadable handwritten field, two similar names that must **not** be merged, one conflicting timestamp, one malformed file

Result: claims become measurable. The team can state accuracy per component with numbers instead of asserting a generic "AI accuracy" figure, which invites challenge and cannot be defended.

8. Consolidated file-change index

PATH	ACTION	PURPOSE
app/extraction/entities_in.py	New	Vehicle, location, organisation, person extractors
app/services/relationship_builder.py	New	Derives typed entity-to-entity edges from records
app/graph/analytics.py	New	Centrality, bridges, communities, paths, subgraphs
app/services/temporal.py	New	Incident windows, sequencing, burst detection
app/services/case_assistant.py	New	Permission-aware, source-citing case Q&A
app/evidence_intelligence/preprocessing.py	New	OpenCV variants and image quality report
app/evidence_intelligence/providers/openai.py	New	OpenAI provider behind the existing protocol
app/parsers/cdr.py, fir.py, surveillance.py	New	Problem-statement source adapters
migrations/versions/*_entity_relations.py	New	The one structural migration
docker-compose.demo.yml	New	Self-contained stack with Postgres and Redis
app/models/entities.py	Modify	EntityRelation ; incident window on Case
app/evidence_intelligence/schema.py	Modify	New record fields; extended SourceType
app/services/entity_resolution.py	Modify	New identity fields and canonicalisation rules
app/graph/connections.py	Modify	Weights for the new entity types
app/alerts/rules.py	Modify	Five criminal-network rules; rule_version
app/services/grounded_pipeline.py	Modify	Invoke relationship builder inside the run boundary
app/api/grounded.py	Modify	Relation listing, review and network endpoints
app/evidence_intelligence/providers/prompts.py	Modify	Request the new entity classes; handwriting states
app/core/config.py	Modify	OpenAI settings; openai_first routing mode
client/src/pages/workspace/*	Modify	Network Intelligence, Entity Profile, Relationship Detail, Timeline
Readme.md	Modify	SIH26189 positioning and honest limitation statement
scripts/generate_synthetic_evidence.py	Modify	Multi-source benchmark case

9. Demo script for evaluation

Plan the build backwards from this sequence. Every phase above exists to make one of these steps possible.

#	STEP	WHAT IT PROVES
1	Create a case; set the incident window	Case-centred workflow
2	Upload six heterogeneous files	Multiple sources
3	Show SHA-256 receipt and preserved original	Evidence integrity
4	Show asynchronous processing stages live	Scalable architecture
5	Show extracted Person, Vehicle, Location, Phone, Organisation	PS entity requirement
6	Review one uncertain handwritten field → "unclear, review required"	Uncertainty handling
7	Resolve one entity across three sources	Cross-source resolution
8	Open the network graph — candidate and verified visually distinct	Relationship maps
9	Click the most important node → read the WHY explanation	Influential individuals
10	Apply the incident-window filter; open the timeline	Temporal intelligence
11	Open one explainable alert with its supporting evidence	Suspicious patterns
12	Click a relationship → claim → evidence → the original file with the region highlighted	The differentiator
13	Ask the assistant a source-cited question	Actionable intelligence
14	Demonstrate case-level access control	Security
15	Show blurry evidence handled safely	Responsible AI

Step 12 is the winning moment. Many teams will show a graph. Very few will be able to click from a graph edge to the exact pixel region of the original document that produced it. The provenance data needed for this is **already stored** in this repository — only the interface is missing. Protect the schedule for Phase 7.

10. Metrics, claims discipline, and risks

Metrics to report

COMPONENT	METRIC
Provenance	Traceability Coverage = graph claims with valid source evidence ÷ total graph claims. Target: 100% , guaranteed by the Phase 2 insertion invariant.
OCR	Character and field accuracy on the labelled benchmark
Entity extraction	Precision and recall, reported per entity type
Relationship extraction	Precision and source-reference correctness
Entity resolution	False-merge rate and possible-match precision
Alerts	Review rate and measured false-positive review rate
Assistant	Citation correctness and permission correctness
Reliability	Model failure rate, retry success, idempotency behaviour

Claims the team must not make

- No claim of guilt, criminality, or crime prediction. The system produces investigation leads.
- No claim of live access to NCRB, CDR, banking, surveillance or intelligence databases. The prototype uses synthetic records; real connectors require lawful authorisation.
- No claim of 100% OCR accuracy, unlimited data, real-time national analysis, or production deployment.
- No generic "AI accuracy" percentage. Report per-component metrics instead.
- Network centrality must be described as **network importance** or **review priority**, never as an indication of guilt.

Risk register

RISK	CONSEQUENCE	MITIGATION
Untracked work is lost	The strongest subsystem disappears	Phase 0, executed first, before any other work
Hallucinated OCR or handwriting	An unsupported investigative fact	Literal transcription, null values, review flags, source boxes — already the codebase's contract
Silent identity merging	A false relationship in the graph	Match scores with explanations; possible-match state; no automatic merge
An edge without evidence	Unusable intelligence; the metric collapses	Insertion-time invariant in Phase 2
Prompt injection inside uploaded documents	Extraction policy subverted	All document text treated as untrusted data; tested adversarially in Phase 9
Scope creep into Neo4j / PaddleOCR / new infrastructure	Schedule lost, demo not ready	Section 5 non-goals are binding decisions

Overclaiming readiness	Loss of evaluator trust	Demonstrate one small complete slice; state limitations plainly
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11. Team allocation and timeline

Total realistic effort is **four to five weeks for a team of four to six**, provided the frontend track runs in parallel with the backend track from week two onwards.

TRACK	PHASES	DEPENDENCY
Track A Core backend	Phase 1 → 2 → 3 → 5	Sequential. Phase 2 unlocks Phases 3, 5 and 7.
Track B Ingestion	Phase 4 → 6	Can start after Phase 1. Independent of Phase 2.
Track C Frontend	Phase 7	Begins against the Phase 2 API contract; does not wait for the implementation.
Track D Assistant & QA	Phase 8 → 9	Phase 9 dataset work can begin immediately and should.

Suggested week plan

WEEK	FOCUS
Week 0	Phase 0 in full. Phase 9 dataset design begins in parallel.
Week 1	Phase 1 complete. Phase 2 schema and migration landed. Frontend agrees the API contract.
Week 2	Phase 2 complete. Phase 3 complete. Phase 4 CDR and FIR adapters underway. Track C building Network Intelligence.
Week 3	Phase 4 complete. Phase 5 complete. Phase 6 preprocessing and routing. Track C building Entity Profile and Relationship Detail.
Week 4	Phase 6 complete. Phase 7 complete. Phase 8 complete. Benchmark run; metrics recorded.
Week 5	Buffer. Rehearse the fifteen-step demo end to end at least three times on a clean clone.

The first three actions

1. **Today** — commit everything, create the branch structure, move secrets out of the repository. One mistaken checkout currently destroys the grounded pipeline.
2. **Tomorrow** — Phase 1 extractors. Most visible gap, least effort, no structural migration.
3. **This week** — the `entity_relations` table and the relationship builder. This is the unlock that makes centrality, the network view and source drill-down possible at all.

Summary in one line. DRISHYAM is an excellent evidence-provenance engine dressed for the wrong problem statement. Do not change the engine. Change the vocabulary, the shape of the graph, and the presentation.

Document status. This roadmap is derived from a direct file-level audit of the `Drishyam_Latest_Integrated` repository, cross-checked against three advisory reports. Where it disagrees with those reports, the reasoning is stated in Section 5. Line references and file paths reflect the repository state at the time of the audit and should be re-verified before implementation begins. Nothing in this document claims that a capability is implemented unless the audit confirmed it in source.